Genome scans and candidate gene approaches in the study of common diseases and variable drug responses

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There is both tremendous enthusiasm and controversy surrounding genetic association studies. In this article we use new methods for representing variation in the human genome that suggest as a few as 170 000 carefully selected single nucleotide polymorphisms (SNPs) are sufficient to represent all the common SNPs in European or East Asian population samples, and up to 300 000 in West African population samples. We also show that efficient genome scans require a detailed description of genomic variation in humans, as envisioned in the HapMap project. Although these analyses are encouraging about the prospects for genome scans, we argue that the difficulty of fine-localization of causal variants will be an important obstacle to progress, and that for reasons of both cost and statistical power candidate gene approaches are likely to remain the primary focus of genetic association studies for years to come.

First we review approaches for carrying out genetic association studies, focusing in particular on strategies, which seek to use patterns of statistical association among single nucleotide polymorphisms (SNPs) to identify a subset of SNPs that efficiently represent the other SNPs in a genomic region, or in the genome as a whole. A SNP or a set of SNPs that have been selected on the basis of linkage disequilibrium (LD) patterns to represent other SNPs are referred to as tagging SNPs (tSNPs) (Box 1).

The identification and evaluation of tagging SNPs

Until recently, most association studies could be described as candidate polymorphism studies [1] in which only one or a few SNPs within a gene of interest were assessed for association with a phenotype. With the arrival of denser SNP maps and cheaper high-throughput genotyping techniques, such studies have given way to true candidate gene studies in which patterns of LD throughout the gene region are determined to (i) select efficient sets of tSNPs in an attempt to represent all the common variation in the gene, and (ii) assess the true weight of evidence for any one highly associated SNP being a functional variant, given that many other SNPs could be in high LD with it. Researchers are now looking to extend this technique to 'tag' the entire human genome [2].

Tagging SNPs were recently associated with the idea that there are often more or less obvious blocks of LD in the human genome [3–5]. The term haplotype tagging SNPs (htSNPs) [3] was used for SNPs meant to represent the major haplotypes in a stretch of limited haplotype diversity. Tagging common haplotypes, however, is only one of many ways of selecting SNPs based on LD properties. In fact, in the context of genetic mapping, it is more relevant to consider the r^2 measure of LD between the tags and the SNPs they are meant to represent [6], which will generally include the functional SNPs that are to be identified. $r^2 = (X_{AB} - X_A X_B)^2/(X_A X_B X_a X_b)$, where X_A, X_a, X_B, X_b, are the frequencies of each of two alleles at two loci, and X_{AB} is the frequency of the two-locus haplotype made up of alleles A and B at the two loci. In an association study, the tSNPs will be typed, and if any of the SNPs that they statistically represent (or tag) are causal (i.e. influence the phenotype), they will drive an association between the tSNPs and the phenotype that will be detectable in a sufficiently large study.

Among the pairwise measures of LD, the r^2 measure is the most appropriate in selecting tSNPs because it allows assessment of the loss of power resulting from typing an associated variant (tSNP) as opposed to the causal variant with which it is associated (see Ref. [6]; Box 1). One approach, therefore, would be to select tSNPs on the basis of their pairwise r^2 values with the tagged SNPs [7]. Although this has some appeal because of its ease of interpretation, there is reason to believe that this approach might not be very efficient. To achieve high r^2 values the frequencies of the variants (i.e. associated and causal) must be matched, and this might be difficult to achieve SNP by SNP, or even between haplotypes defined by the tSNPs and the other SNPs, thus requiring an unacceptably large number of tSNPs (Box 1).

An alternative approach focuses on the coefficient of determination (i.e. haplotype r^2) in a linear regression, which uses the haplotypes defined by the tSNPs to predict the state of the tagged SNPs (http://popgen.biol.ucl.ac.uk/software.html and http://www-gene.cimr.cam.ac.uk/clayton/software) (Box 1). The intuition behind this approach is that even when individual haplotypes

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http://tigs.trends.com 0168-9525/$ - see front matter doi:10.1016/j.tig.2003.09.006

Box 1. Identifying and evaluating tagged single nucleotide polymorphisms (SNPs)

Here, we consider the related issues of how to select single nucleotide polymorphisms (tSNPs) and how to evaluate how well they are expected to perform in representing, as yet, undected SNPs.

As discussed in the main text, the most direct approach is to identify a set of tSNPs that maintain high r^2 values directly with the other (tagged) SNPs (or that define a set of haplotypes that do so). r^2 values can be defined as coefficients of determination (i.e. the proportion of explained variation) obtained from fitting linear models predicting the state of the tagged SNP from the set of tSNPs. For haplotype r^2, the model is fitted to the G haplotypes defined by the tSNPs:

$$Y_i = x_{i1}b_1 + x_{i2}b_2 + \ldots + x_{iG}b_G \qquad \text{[Eqn I]}$$

Where Y_i is the predicted state of the tagged SNP of interest on the ith chromosome (e.g. representing observed values as 0 or 1), $x_{i1} \ldots x_{iG}$ are indicator variables (0 or 1 depending on which haplotype is present) and $b_1 \ldots b_G$ are coefficients estimated by standard least squares from the observed data. Pairwise r^2 values between two SNPs (or between a SNP and the presence or absence of a single haplotype) can be defined by setting $G = 2$. If the tagged SNP state or the haplotype state on chromosome i are not known with certainty, weighted least squares can be used, either by splitting each observed chromosome into all possible SNP and haplotype states with weights equal to their estimated probabilities, or equivalently by listing all SNP and haplotype combinations consistent with the data and using weights equal to their estimated population frequencies (e.g. from EM estimation). Software for the fitting and evaluation of r^2 values is available in the TagIT package (http://popgen.biol.ucl.ac.uk/software.html).

Pritchard and Przeworski [6] showed, under simplifying assumptions and when $G = 2$, that the power of a case-control study comprising n observations looking directly at a causal SNP was equivalent to the power of a case-control study comprising n/r^2 observations looking at a marker, where r^2 is the coefficient of determination between the marker and the causal SNP. We have verified this simple power relationship using simulations, and have also found that the relationship holds when $G > 2$ (M.E.W., unpublished).

To illustrate these approaches, we identified four tSNPs for region 25b in the Gabriel *et al.* [13] dataset using different criteria. The 4 tSNPs for the European sample provide a haplotype r^2 of 0.96 or above for each of the seven 'tagged' SNPs in this region. Many of the tagged SNPs in this region, however, fail to show high pairwise r^2 values with any single SNP. For example, the tSNPs are SNPs 4, 5, 8 and 11 in order through the region. For tagged SNP 1, however, the pairwise r^2 values against the tSNPs are 0.32, 0.08, 0.02, and 0.05 (Figure I). Similarly, the pairwise r^2 values between the haplotypes defined by these tags and the tagged SNPs do not perform better; they are all below our cut-off of 0.85. These comparisons suggest that in the selection of tagging SNPs, and in the subsequent association study, it might be necessary to consider association tests that either use a regression model with haplotypes as independent predictors or employ other approaches which consider clusterings of haplotypes (e.g. [29]).

After selecting a set of tSNPs, it is necessary to assess how well they represent as yet undetected SNPs. We used the terminology of Weale and Goldstein who defined K as the total number of SNPs identified in the region of interest, H as the tSNPs identified from K, and A the set of all common SNPs in the region (http://popgen.biol.ucl.ac.uk/software.html). The question to answer is how well does H represent the set of SNPs in A. To assess this, we have used a sub-sampling procedure in which we sequentially drop out SNPs from the set K and identify a set of tSNPs in the reduced set. We then checked the performance of the tSNPs in predicting the state of each of the dropped SNPs in turn, and so estimate the performance against the (unknown) elements of A. There are, two potential problems to this approach. (i) There is a causal SNP of interest in the region, but its linkage disequilibrium (LD) properties are not randomly drawn from A. This could result from selection for (or against) the causal SNP [11]. (ii) The LD properties of K are not randomly drawn from A. In particular, the spatial distribution of K in A will not be important in a completely 'block'-like LD region, but, if incomplete LD exists and certain sub-regions are under-represented in K then this will be important. How important these problems are in practise remains to be resolved.

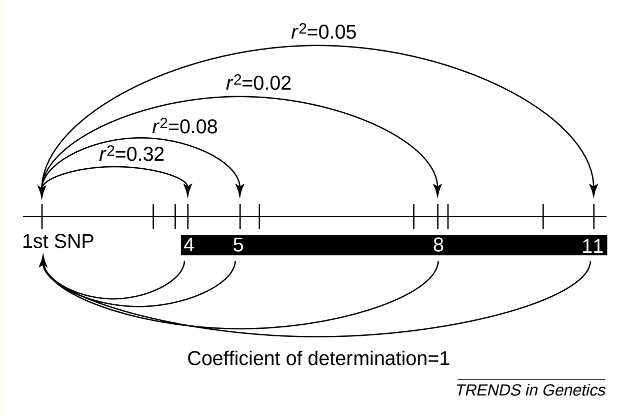

Fig. I. Illustration of relative SNP positions and pairwise r^2 values for the 11 SNPs in region 25b of the Gabriel *et al.* [13] dataset, as described in Box 1. The tSNPs for this region are SNPs 4,5, 8 and 11. As shown, none has a high pairwise r^2 against SNP 1, although SNP 1, like the other SNPs, is well predicted by the haplotype r^2 criterion. Abbreviations: SNP, single nucleotide polymorphism; tSNP, tagging single nucleotide polymorphism.

defined by tSNPs do not correlate perfectly with tagged SNPs, combinations of the haplotypes might do so, and these combinations are identified by selection of the appropriate coefficients in the linear regression. In practice, we have found that in many cases the tSNPs (or the haplotypes they define) might fail to represent other SNPs even though the coefficient of determination is 1 using the haplotype r^2 criterion (Box 1).

Blocks of linkage disequilibrium?

Although inspired in part by the idea of blocks of LD, the tagging approach does not require clear blocks of LD. Regardless of how discrete the pattern of LD is, or what causes this pattern, it is the predictability of the state of tagged SNPs by the haplotypes defined by the tSNPs that determines power in a genetic association study. Thus some of the controversies surrounding the definition and the underlying causes of LD blocks [8] are not relevant to the use of tSNPs. Indeed, Zhang *et al.* [9] have shown with simulations that even with recombination rates that are uniform throughout the genome, genotyping ~25% of SNPs would result in only negligible loss of power. The magnitude of savings provided by tags will, however, depend on the extent to which recombination is localized in hotspots, which remains unknown, and also the demographic history of the population under study [10]. The properties of tSNPs, therefore, can only be accurately assessed empirically.

It is already clear, however, that some genomic regions will be considerably harder to tag than others, and that tSNPs will behave somewhat differently in different population groups. It is also possible that natural selection could make tSNPs perform less well in practice than predicted. For example, positive selection could elevate the

Table 1. Genome-wide tags[a]

Chromosome	Region[b]	Size (kb)	No. SNPs[c,d]	Density[e]	Number of tSNPs Asia	Number of tSNPs Europe
2	8a	163	17 ∣ 21	10 ∣ 8	9	8
2	9a	147	28 ∣ 15	5 ∣ 10	8	8
2	11a	94	25 ∣ 22	4 ∣ 4	5	7
2	12b	24	18 ∣ 9	1 ∣ 3	5	3
6	26a	170	27 ∣ 39	6 ∣ 4	8	11
8	30a	178	17 ∣ 40	10 ∣ 4	9	8
9	31a	175	14 ∣ 9	12 ∣ 19	7	3
9	31b	102	18 ∣ 25	6 ∣ 4	7	7
11	36a	192	24 ∣ 28	8 ∣ 7	6	5
13	39a	172	24 ∣ 32	7 ∣ 5	7	7
15	43a	157	26 ∣ 18	6 ∣ 9	8	9
16	44a	88	20 ∣ 28	4 ∣ 3	8	7
22	52a	237	22 ∣ 32	11 ∣ 7	9	9
Total	NA[f]	1899	285 ∣ 318	NA[f]	96	92
Mean	NA[f]	146	22 ∣ 24	7 ∣ 7	7	7

[a]The number of single nucleotide polymorphisms (SNPs) required to tag the human genome for the Asian and European samples are ~170 000 and 150 000, respectively. These calculations are based on the assumption that the human genome comprises 3290 megabase pairs of DNA. We note the results do not provide a direct assessment of power in representing SNPs that are not yet discovered because the tagging SNPs (tSNPs) are selected on the basis of the known SNPs (Box 1). We have however evaluated the European dataset using a re-sampling approach described in Box 1 and TagIT and find that using this assessment the expected power is similar to that show in Figure 1 (Figure 2), for all regions except 9A and 31A. Excluding these two regions where the original density of SNPs appears to have been insufficient, the estimate for the number of tSNPs required for the whole genome for the European sample rises to 170 000. The re-sampling approach to assess sufficiency of the original genotyping data was not applied to the other population samples.

[b]Only those regions with a total number of 10–30 SNPs in the Asian dataset were used in the analyses. Note that this does not mean that the European regions also have between 10–30 SNPs in each region because this number varies across the populations. This ascertainment scheme was used because regions with too few SNPs would create a biased estimate (resulting in the selection of all SNPs present as tags) whereas regions with more than 30 SNPs presented computational complications for phase inference and tagging analyses using TagIT. The density for the 13 regions analyzed here is comparable to the density for all the regions from Gabriel *et al.*

[c]The slash ∣ divides values in Asia and Europe for number of SPNs and density columns.

[d]The number of SNPs with minor allele frequencies ≥8% for each region.

[e]The density is given as the number of kb per SNP in the region. The correlation between density of SNPs and number of tSNPs is 0.57 ($P = 0.04$) and −0.26 ($P = 0.31$) for the Asian and European populations.

[f]Abbreviation: NA, not applicable.

frequency of recent, causal mutations, resulting in young but high frequency haplotypes. Weale *et al.* [11] have shown that the tagging strategy might have difficulty with such discrepant SNPs because if such a SNP is not included in the first screen of variation, the other SNPs in the region are unlikely to fully represent it. If SNPs that influence drug response and disease predisposition are more likely to have experienced such selection, this could pose significant complications for haplotype mapping.

Genome-wide tags

Exhaustive strategies for finding functional variants that attempt to itemize all the SNPs that might be of functional significance (e.g. all coding SNPs, or all SNPs in and near a gene, as in Ref. [12]) are difficult to implement on a genome-wide scale. These strategies depend on a prior knowledge of the genomic location of all the important functional elements of the genome, which remains well out of reach. Hopefully, genome-wide LD scans will offer an efficient alternative, and two directions have developed in this field. The first is to genotype a set of markers spaced evenly throughout the genome (i.e. spaced SNPs), not optimized for any particular tagging features (e.g. http://www.sequenom.com). The second is to identify a set of genome-wide tSNPs. The tSNP approach requires an assessment of genome-wide LD patterns, which is the aim of the recently launched HapMap project [2]. Here, we compare the two methods, tSNPs and spaced SNPs; beginning with an estimate of the number of tSNPs required for whole genome scans.

Gabriel *et al.* [13] recently showed that there are many tracts of low haplotype diversity in the human genome, and estimated the number of SNPs required to tag the genome by calculating the number required to represent the estimated number of common haplotypes in the genome. These results are generally encouraging for genome-wide association studies, but they do not permit direct assessment of the number of tSNPs required to have a certain level of statistical power. Here, we provide an alternative estimate of the number of tSNPs required to tag the human genome using the haplotype r^2 criterion for tag selection (implemented in the TagIT software package (http://popgen.biol.ucl.ac.uk/software.html).

For 13 of the regions reported in Gabriel *et al.,* we found that 96 and 92 tSNPs are sufficient to tag the studied genomic regions in the Asian and European datasets, respectively (Table 1). We have also carried out a comparable analysis of the African dataset, but subdividing some of the regions as a result of computational constraints. We found that 173 tSNPs are required. The subdivisions, however, systematically increase the number of tags estimated, so this should be viewed as an upper bound on the number of tSNPs necessary for these regions (see below).

The performance of these tSNPs in representing the tagged SNPs is very encouraging. The vast majority of SNPs are predicted perfectly by the tSNPs using the haplotype r^2 criterion (Figure 1). The 13 regions studied represent a total of 1899 kb or ~0.06% of the human genome. Assuming that these different regions have (parametrically) different LD properties, then the appropriate way to estimate the number of tSNPs for the entire genome with these results is to take the average of the number of tSNPs per kilobase across the regions and then

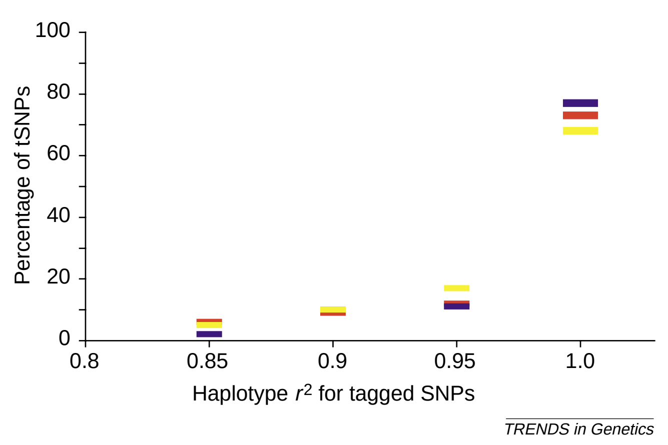

Figure 1. Distribution of haplotype r^2 values for the 'tagged' SNPs compared with the haplotypes defined by the tSNPs. As shown a majority of the tagged SNPs [72%, 76%, and 68% for Asia (red), Europe (blue) and Africa (yellow)] are completely tagged; the haplotype r^2 using haplotypes defined by the tSNPs to predict the tagged SNPs gives a coefficient of determination of 1.0 (see Box 1 for a description of the tSNP selection criterion). Abbreviations: SNP, single nucleotide polymorphism; tSNP, tagging single nucleotide polymorphism.

scale this up for the size of the entire genome. This approach leads to an estimate of ~170 000 SNPs for the Asian sample and 150 000 for the European sample (also see below), and 300 000 for the West African population, ~50% of the previous estimates for the Asian and European samples, and <33% for the Africans using the same data [14].

Will the tSNPs represent non-identified variants?

The analyses above demonstrate that there is considerable redundancy among SNPs, and that the haplotype r^2 criterion efficiently capitalizes on this redundancy to identify a minimal set of tSNPs. But the question remains whether selected tSNPs will adequately represent SNPs that have not yet been identified. As has been noted previously, if the original dataset is not sufficiently dense then tSNPs are unlikely to represent as yet non-identified SNPs, even if they do represent the SNPs already known [11,14].

We have addressed this question by doing 318 subsampling experiments on the European dataset, dropping each in turn one of the 318 SNPs (Box 1). For each experiment, we then identified a new set of tSNPs for the appropriate region, equal in size to original set, but ignoring the dropped SNP in the selection. We then checked the haplotype r^2 for the new tags against the dropped SNP. In general, the dropped SNPs are very well predicted by the tSNPs that are selected without considering the dropped SNP (Figure 2). In particular, the average haplotype r^2 for the 318 SNPs is 0.944 compared with 0.998 in the original analysis. Even more encouraging, excluding the two least densely typed regions (31a and 9a with one SNP every 19 and every 10 kb, respectively), there was little or no correlation between haplotype r^2 for the dropped SNP and the density of SNPs typed (Figure 2). If we exclude these two regions where the original density of SNP genotyping appears not to have been sufficient, the new estimate for the number of tSNPs required for Europe increases to 170 000. We also note

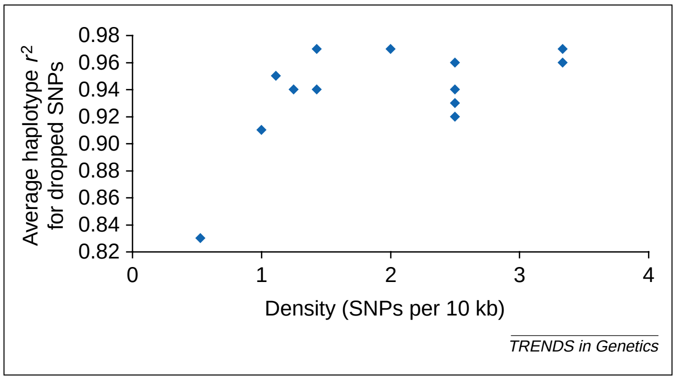

Figure 2. Average haplotype r^2 as a function of the original density of SNP genotyping for each of the 13 regions analyzed in Table 1. For every region, each of the SNPs was dropped in turn and tSNPs were identified. The haplotype r^2 for the dropped SNP was then determined. The average for each region is reported against the density, showing that the performance improves until ~1.5 SNPs per 10 kb, but no further increases are apparent after this point. Abbreviations: SNP, single nucleotide polymorphism; tSNP, tagging single nucleotide polymorphism.

there was no correlation between the haplotype r^2 for the dropped SNP and its frequency for the SNPs analyzed here (i.e. minor allele frequency >8%). It will be important to assess this relationship for SNPs with minor allele frequencies below 8%. We conclude that, in general, the average marker density necessary to find a set of tags sufficient to represent the common allelic variants in the human genome is >1 per 10 kb, but <2 per 10 kb, although we do expect that a small number of SNPs with very unusual LD properties would still be missed (Box 1). Note that these results are not necessarily in conflict with the claim that the length of haplotype blocks decreases with increasing marker density [14]. Depending on how a block is defined, a rare recombinant haplotype might 'break a block', but have little effect on whether the selected tSNPs can adequately represent the other SNPs present. Thus, we would argue that although blocks of LD have provided a very useful illustration of the limited haplotype diversity in the human genome [13] the identification of block boundaries, however they are defined, do not provide an appropriate guideline for selecting tSNPs. Thus, the density or marker typing necessary to achieve stable block definitions will not in general coincide with the density of markers necessary to select a set of tSNPs with desirable properties for genetic association studies, as outlined here. Although larger datasets will be necessary to confirm our conclusions, our analyses strongly imply that the density used in the Gabriel dataset is sufficient for the identification of tSNPs that would represent the majority of common polymorphisms in the human genome, even if not sufficient for the delimitation of block boundaries. These results have obvious relevance to experimental design in the HapMap project.

Do tSNPs perform well in a different sample from the same population?

Another concern with the selection of tSNPs is whether the original population sample is sufficiently large to define tSNPs that will work well in another sample from the same population. This is of particular concern when large or low LD regions are analyzed, and haplotype frequencies are

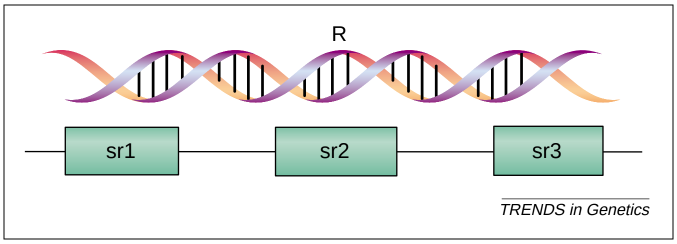

Figure 3. Two approaches that could be used to identify a best set of tSNPs to be used in an association studies. The total number of tSNPs needed to tag the entire region R is equal to S_R, whereas the total number needed to tag the 3 sub-regions sr1, sr2, and sr3 is equal to $S_{sr1} + S_{sr2} + S_{sr3}$. Analyses suggest that in the human genome associations are often long range, so that the number of SNPs required for the region R is often much less than the sum of sr1, sr2 and sr3 (i.e. $S_R \leq S_{sr1} + S_{sr2} + S_{sr3}$). Abbreviations: SNP, single nucleotide polymorphism; tSNP, tagging single nucleotide polymorphism.

consequently low. To address this question, we constructed two bootstrap resamples of individuals for regions 8a and 9a. For each region, we then selected tags for one of the resampled populations, and applied these tags to the other resampled population. We found the tags worked almost as well, with the weighted haplotype r^2 being reduced by $<5\%$, and at most one tagged SNP falling below the threshold of 0.85. This suggests that, for regions like those studied here, 100 chromosomes might generally be a sufficient sample size for the selection of tSNPs. Preliminary analyses suggest, however, that ~50 chromosomes are not sufficient (results not shown).

Why so few tSNPs?

The difference between our estimate and that of Gabriel *et al.* appears to depend primarily on the inclusion of long-range associations in our identification of the tags (Gabriel *et al.* had already noted such associations would reduce the required number of tSNPs). For example, consider selecting tags within each of three consecutive sub-regions (denoted sr1, sr2, sr3) separately, or for the three regions together (denoted R; Figure 3). Denote the number of SNPs required to tag R as S_R, and the total number required for the sub-regions as $S_{sr1} + S_{sr2} + S_{sr3}$. Owing to long-range LD, $S_{sr1} + S_{sr2} + S_{sr3} \geq S_R$. Although the quantitative effect will depend on the definition of sub-regions, various divisions of the available data make clear that the difference can be substantial (Table 2). We note here that even in our analyses the numbers will be inflated by the effects of dividing genomic regions, and we expect that when larger contiguous datasets are analyzed the estimate number of tSNPs will drop further.

The results of Table 2 show that tagging regions can be more efficient than tagging sub-regions. However, in the context of candidate gene studies, it would be preferable to have optimized tags for specific regions. If tags have been selected for larger regions, they will not necessarily include a subset that is optimal for a specific region (e.g. candidate gene) of interest. The difference in SNPs required to tag regions and sub-regions, even in the simple comparison of Table 2 suggests that region-wide SNPs might often be much less efficient at representing variation in sub-regions. This trade off might need consideration in the context of the HapMap project, and would be facilitated by the availability of a set of high-quality candidate gene studies that could serve as benchmarks against which to compare different approaches for tSNP selection in the HapMap project. This consideration therefore argues against waiting for the completion of the HapMap project before the community continues with candidate gene work.

Even-spaced SNPs

To evaluate the even-spaced strategy, we first used the same number of SNPs as required to tag a region, but we distributed the SNPs evenly through the region without regard to LD patterns. For five of the regions from the data of Gabriel *et al.*, we evaluated the weighted haplotype r^2 and the minimum haplotype r^2 (http://popgen.biol.ucl.ac.uk/software.html) for these 'even-spaced' SNPs in predicting the allelic states of the other SNPs in the studied regions (Table 3). In all cases, the tSNPs performed substantially better than the spaced SNPs in terms of both the weighted average haplotype r^2 and the number of SNPs missed (i.e. SNPs not tagged well). For example, using spaced SNPs, 29% of the unselected SNPs (i.e.18 SNPs) had a haplotype $r^2 < 0.85$ with the haplotypes defined by the spaced SNPs.

We also evaluated a much denser evenly spaced map, with markers approximately every 10 kb (Table 4). In this case, the spaced SNPs require a total of 47 SNPs in comparison with 30 tSNPs. Even with 50% more SNPs, however, the spaced SNPs perform worse in several ways. For example, they result in four of the unselected SNPs with a haplotype r^2 value below 0.85, whereas the tagging strategy has none. These observations are not a criticism of current 'even-spaced' strategies (e.g. [15]) in that genome-wide tSNPs are not currently available. Moreover, a very dense set of even-spaced SNPs is expected to represent many other SNPs. The analyses show, however, that current even-spaced strategies should not be viewed as

Table 2. Tagging regions and tagging sub-regions[a]

				Number of tSNPs for each sub-region	
Region	**Size (kb)**	**Total number of SNPs**	**Number of tSNPs for each region**	**Sub-region 1**	**Sub-region 2**
8a	163	21	8	6	6
11a	94	22	7	4	5
36a	192	28	5	4	4
44a	88	28	7	5	7
Total	537	99	27 tSNPs required	41 tSNPs required	

[a]The number of tagging single nucleotide polymorphisms (tSNPs) required to tag four genomic regions from the European dataset reported in Gabriel *et al.* [13] in comparison to sub-regions that divide the total region into sub-regions of approximately equal size. The precise criteria used to select tSNPs for each region and sub-region were that the overall weighted r^2 was >0.95 and that the minimum r^2 between the tags and the tagged SNPs was >0.85; (i.e. all tagged SNPs are represented to a very high degree). In all cases the full-region approach is more efficient than tagging individual sub-regions as a result of the presence of significant LD across sub-regions.

Table 3. Comparing the power of tSNPs and evenly spaced SNPs[a]

Region	Size (kb)	Number of SNPs	Number of tSNPs	Density (kb)[b]	Weighted r^2	Minimum r^2 [c]
11a	94	22	7	13.4	0.97	0.78 (1)
12b	24	9	3	8	0.92	0.84 (4)
31b	102	25	7	14.6	0.70	0.26 (11)
36a	192	28	5	38.4	0.94	0.57 (2)
Total	412	84	22	NA[d]	NA[d]	NA (18)
Mean	103	21	6	18.6	0.88	0.61 (5)

[a]The results are for an 'even-spaced' strategy where the number of even spaced single nucleotide polymorphisms (SNPs) is equated to the number needed to 'tag' the region. The weighted haplotype r^2 and the minimum haplotype r^2 for the even spaced SNPs and all other SNPs in the studied region are reported.
[b]Density is reported as the number of kb per SNP (i.e. size in kb divided by the number of spaced SNPs.)
[c]The numbers in brackets represent the number of SNPs not tagged for each region, where not tagged is defined as haplotype $r^2 < 0.85$
[d]Abbreviation: NA, not applicable.

comprehensively representing common variation in the human genome. More importantly, these analyses make a compelling case that the HapMap project will permit substantially more efficient genome-wide screens.

Candidate genes versus whole-genome scans

Despite these encouraging points for genome scanning, we recommend that association studies should generally begin with candidate genes and regions (e.g. linkage supported). Ignoring genotyping costs, scanning the genome as a whole still entails a high statistical price. For example, tagging a pathway relevant to a condition of interest (e.g. renin-angiotensin pathway and hypertension) might require ~200 tSNPs, in comparison with >100 000 for the whole genome. Assume that there is one causal SNP for hypertension in this pathway with a minor allele frequency of 0.1 and an odds ratio of 1.5. Assuming tSNPs are grouped into sets of five (that are optimized to represent other SNPs in the relevant region) and that the causal SNP has a haplotype r^2 of 0.85 against just one such set of tSNPs, then re-arrangement of the formulae in Johnson *et al.* [3] and Pritchard and Przeworski [6] shows that a study with 1000 cases and 1000 controls would have a 32% chance of finding the causal SNP examining only the 200 tSNPs for the pathway. For a genome scan, taking the very conservative number of 100 000 SNPs required, we would have only a 2.6% chance of detection because of the need to correct for separate tests of 20 000 sets of tSNPs.

Of course, the real statistical price of typing more markers depends on how the analyses are done including the trade-off accepted between false positive and false negative error rates. More generally, it might be preferable to move away from this standard frequentist approach to using prior information about the likelihood that a genome region carries a causal variant in a Bayesian framework. For the moment, it seems to us that it would be difficult to make distinctions on prior probability much more finely than establishing one level for a candidate gene or region (e.g. linkage supported), and another lower level for the rest of the genome, and thus it seems sensible to concentrate energies on the candidate areas.

Selecting candidate genes

Although candidate gene lists will rarely, if ever, include all relevant genes, it is also true that some parts of the genome are better places to start looking than others. Candidate genes are particularly obvious in the case of variable drug response, for example, genes encoding metabolizing enzymes and transporters, or targets and associated pathways. Indeed, one striking feature of pharmacogenetics is how often relatively 'obvious' candidate genes turn out to carry important variants [1,16–17].

Although candidate genes for disease susceptibility are less obvious, there are many useful pointers. For many complex diseases, genes have been identified that cause rare, mendelian forms of the conditions. It is reasonable to assume that genes that have severe mutations causing mendelian forms might also harbor less severe mutations that predispose to disease, and indeed some of these have been implicated in more generalized risk as, for example, the *Tau* gene in parkinsonian conditions [18]. These mendelian genes constitute high priority candidate genes for the relevant conditions, and deserve the systematic approach described here.

Table 4. Comparing the power of tSNPs and evenly spaced SNPs[a]

Region	Size (kb)	Total number of SNPs	Number of tSNPs[b]	Weighted r^2 [c]	Minimum r^2 [d]
11a	94	22	9 (7)	0.97 (0.99)	0.49 (**2**)
26a	170	39	15 (11)	0.96 (0.98)	0.80 (**1**)
31b	102	25	9 (7)	0.96 (0.95)	0.80 (**1**)
36a	192	28	14 (5)	0.99 (0.96)	0.91 (**0**)
Total	558	114	47 (30)	NA[e]	NA[e] (4)
Mean	139	28	12 (7)	0.97 (0.97)	0.75 (1)

[a]The results from the analyses where single nucleotide polymorphisms (SNPs) every 10 kb were selected as tagging SNPs and the weighted r^2 and the minimum r^2 for these tSNPs and all other SNPs in the studied region were noted. The numbers in brackets represent the corresponding values when a tagging approach is used.
[b]The number of even spaced SNPs is indicated, and the number of tSNPs in parentheses. The number of SNPs used in the 558 kb of the human genome using the tagging framework and the even-spaced framework is 30 and 47, respectively. 47 departs from the expected number of 55 because of difficulty in placing SNPs at precise t10 kb intervals.
[c]The number of regions tagged better in terms of weighted haplotype r^2 by the tagging approach (in parentheses) is 2, whereas the number of regions tagged better by the even-spaced approach is also 2.
[d]The minimum haplotype r^2 for the even spaced SNPs. The number of SNPs not tagged is shown in bold, that is, having a haplotype $r^2 < 0.85$, which is four SNPs for the even-spaced and none for the tagging strategy.
[e]Abbreviation: NA, not applicable.

A second tier of candidate genes often emerges from ideas concerning the biology of the conditions, based on the mendelian forms, animal or cellular models, or the pathophysiology of the condition. For example, the biology of epilepsy suggests channel genes as obvious candidates, whereas the 'xenobiotic theory' [19] suggests that subtle variation in metabolizing profiles among individuals influences susceptibility to Parkinson's disease (PD). A third group of candidates emerges from linkage studies. In this case the candidates are genomic regions often larger than a mega base in size, but the tagging approach applies in the same way as for candidate genes.

Health warnings for association studies

It should be accepted that association studies are inherently prone to inaccuracy and even abuse. Some of the problems are well documented, such as reporting biases, population stratification, and the need to correct for multiple comparisons across SNPs or haplotypes [20–22].

Other problems seem less well appreciated, such as choosing significance thresholds for multiple tests performed on partially correlated phenotypic data and the difficulties of ultimately making diagnostic use of markers that have been associated to phenotypes, but that are probably only surrogates for the underlying causal variants. For example, the diagnostic use of an associated marker or markers would be of concern in any group that has a different ethnic composition from the one in which the associations were first reported. This makes, for example, the use of genome-wide profiles to predict adverse drug reaction problematic because these will almost certainly draw upon associations between the markers and the unknown causal variants. The use of these associations diagnostically will require explicit evaluation of how well they can be transferred across different population groups. We also note that the presence of long-range LD means that causal variants will often be at a considerable distance from tSNPs that represent them, making it difficult to track down association with a set of tSNPs.

The value (and reputation) of association studies would be much improved if there were an agreed set of minimum standards. As outlined in Goldstein [1], these should currently include a check for stratification [21,23] for all reported associations, although this requirement could be relaxed in time in some settings because stratification appears to be modest in practice (e.g. [24]). Relatively smaller case-control studies searching for variants of modest to high effects would be less sensitive to stratification than larger studies seeking variants of weak effect. In all cases, however, the pattern of LD around associated variants should be assessed to, at least roughly, determine the associated interval [1]. Within the associated interval, efforts should be make to identify all the variants that have high LD with the associated variant. The common occurrence of relatively long-range LD suggests that this will often be more difficult than seems generally appreciated [25]. Once all variants in the interval have been identified, they will provide a set of candidate causal variants whose functional effects should then be assessed, where possible, *in vitro* and/or *in vivo* as appropriate.

Conclusions

Ultimately, the importance of genetic association studies will depend on the variants that predispose to disease and/or influence drug response. At present, very few risk-conferring alleles for common diseases are known. It is likely that haplotype mapping will identify some new common variants of at least modest effect in addition to the ten that have been securely identified [20,26], although it remains possible that these will represent a small fraction of the predisposing alleles for common disease. We will not know, however, until haplotype mapping has been systematically implemented for several common diseases. We would note that those worried that genetic association studies would be of limited value because of the rarity of common variants that influence common disease [27,28] should be re-assured. Although we cannot be confident that the genetics of disease predisposition will prove generally tractable in the context of genetic association studies, there is also strong evidence concerning the genetics of variable drug response. Functionally important variants of high allele frequencies have now been identified in genes encoding drug metabolizing enzymes, drug transporters, drug targets and associated pathways [16–17]. We expect that association studies will identify a great many more such pharmacogenetic variants in the coming years, and will help to track down which precise variant(s) are responsible in the implicated genes. The promise of the human genome project to deliver practical benefits in treating common medical conditions might therefore be realized first through pharmacogenetics as opposed to the genetics of common disease.

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